3. Rotating field system has two slip rings only and it can be easily insulated.
4. Rotating field is light in weight and hence it can run at high speed.
5. In stationary armature the winding may be cooled more efficiently.

8.2 BASIC PRINCIPLE

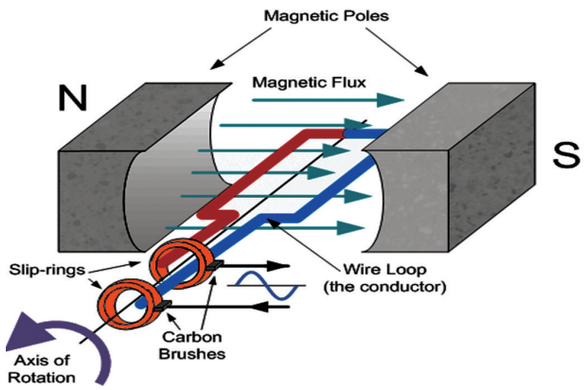


Fig 8.2 *Basic Principle of Alternator*

An alternator works on the same fundamental principle of electromagnetic induction as D.C. Generator i.e. when the flux linking a conductor changes, an emf is induced in the conductor. Like a D.C generator, an alternator has an armature winding and field winding.

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The frequency of output ac voltage of a synchronous generator is directly proportional to the rotor speed. To maintain constant frequency, the rotor must always move at synchronous speed.

8.3 CONSTRUCTION OF ALTERNATOR

Alternators are constructed in two types

1. Salient Pole alternator
2. Non – Salient Pole alternator (or) Turbo Alternator

8.3.1 Salient Pole Alternator

(a) STATOR

It is the stationary part of the machine and it is built up of silicon steel laminations having slots on its inner periphery to house the armature conductors. The armature core in the form of a ring is fitted to a frame which may be of cast iron or welded steel frame which is called stator frame.

The armature core is laminated to reduce the eddy current loss. The laminations are stamped out and insulated from each other with paper or varnish. The stampings also have holes which make axial and radial ventilating ducts to provide efficient cooling.

Armature Slot

Slots provided on the stator core are mainly of three types.

1. Wide-Open Slots
2. Semi-closed slots
3. Closed slots

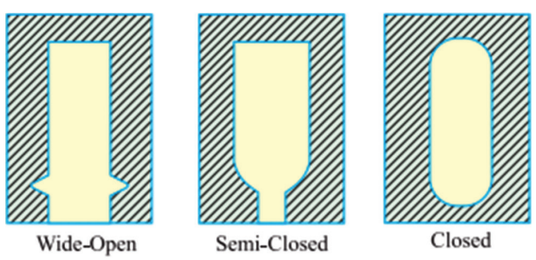


Fig 8.3 *Types of Slots*

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